Trust-Us-Banking Core Engine

Functional & Technical Specifications Document

Cahier des Charges Functional Scheme

Project Team Assignment Blueprint

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Level 3 Computer Science Practical Assignment

1 Project Overview & General Architecture

The **Trust-Us-Banking** core system is an API-driven decentralized central clearing network designed to bridge traditional commercial banking nodes (e.g., UBC, AFB, ECO) with localized digital mobile money wallets (representing MoMo or Orange Money frameworks).

The system architecture focuses on extracting identity control away from physical branch limitations. A single, unique global profile handles multi-ledger assets concurrently, ensuring clean data tracking, high-speed transactional logic processing, and localized verification mechanisms.

2 Role-Based Access Control (RBAC)

The system enforces a strict division of access parameters based on two primary user profiles:

- **Bank Administrator (ADMIN):** Holds complete system visibility, data management control overrides, and exclusive authority to restore restricted client profiles to functional operational states.
- **Client User (CUSTOMER):** Restricted to their specific asset views, personal wallet interactions, security workspace parameters, and outbound transfer management tools.

2.1 System Permissions Matrix

The data lifecycle management layout utilizes the following permissions configuration map:

Table 1: System CRUD Authorization Boundaries

Module Target	Create (POST)	Read (GET)	Update (PUT)	Delete (DELETE)
Banks Matrix	Admin Only	Authenticated	Admin Only	Admin Only
User Registry	Open Register	Admin / Self	Admin / Self	Admin Only
Ledger Accounts	Auto-Trigger	Admin / Self	Admin Only	Admin Only

3 Detailed System Functionalities

3.1 Module 1: Bank & Profile Management (CRUD Layout)

Feature 1: Bank Management (CRUD)
Provides the central control mechanisms required to manage commercial banks within the system. Administrators can register new entities using unique parameters (e.g., Name: "United Bank of Cameroon", Code: "UBC"), update branch details, view active paths, or remove an inactive node entirely.

Feature 2: Global Registration & Matricule Automation
Collects a user's Name, Email, Phone Number, Password, User Type, and initial target Bank Code. Upon processing, the backend automatically generates a unique tracking sequence identifier (**Matricule**), provisions a global master profile, and sets up their first checking account.

Feature 3: Existing User Multi-Bank Enrollment
Enables an existing user to open secondary bank accounts across different banking nodes instantly. The customer only needs to provide their existing Matricule and the new Bank Code; the system verifies their profile and sets up the new account without altering their underlying mobile wallet.

3.2 Module 2: Digital Wallet Infrastructure

Feature 4: Automated Wallet Provisioning & Welcome Cash
Automatically runs in the background immediately after global registration finishes. It creates a dedicated **Local Wallet Account** linked to the user's Matricule, maps their physical mobile phone number as the searchable account identifier, and preloads it with a starting baseline balance of **100,000 FCFA**.

3.3 Module 3: Security, Sessions & Gateways

Feature 5: Bank-Prefixed Logins & Workspace Routing
To enter the application, users input their Matricule and a password prefixed by a valid target bank code (e.g., `UBC_myPassword123`). The engine splits this prefix, verifies credentials, and immediately opens up that specific bank workspace dashboard for the user session.

Feature 6: 3-Strike Lockout Constraint & Administrative Reset
Tracks consecutive failed login attempts on an account profile. On the 3rd consecutive failure, the profile state flips to `BLOCKED`, rejecting all further attempts. The account remains locked until a Bank Administrator executes a reset routine, clearing the counters back to an active `LOGOUT` state.

Feature 7: 30-Minute Automatic Session Timeout
A background countdown clock starts at successful login. If a user session remains active or idle for exactly 30 minutes, the authentication token expires, terminating the session and forcing the user back to the login screen.

3.4 Module 4: Transaction Processing Ledger

Feature 8: Self-Service Internal Deposits & Withdrawals

Provides the core financial bridge between a user's multi-ledger assets:

- **Self-Deposit:** Shifts money out of the user's phone-linked Local Wallet and places it directly into their account at the active context bank.
- **Self-Withdrawal:** Deducts funds from their active bank account balance and pushes it instantly back into their phone-linked Local Wallet asset pool.

Feature 9: Peer-to-Peer Transfers with Percentage Fee Tiering

Routes capital securely from the sender's active bank account directly to another user's bank account anywhere in the network using the recipient's Matricule. Fees are calculated dynamically based on the routing path:

- **Intra-Bank Fee (0.5%):** Applied if both accounts reside within the same bank entity.
- **Inter-Bank Fee (1.5%):** Applied if the transaction moves across separate banking entities.

3.5 Module 5: System Auditing & Transaction Resilience

Feature 10: "All-or-Nothing" ACID Protection Constraint

Ensures absolute transaction safety. All steps in a deposit, withdrawal, or transfer are grouped together as an indivisible unit of work. If any step fails (e.g., a connection drop or insufficient funds), the entire operation cancels instantly, reversing any changes so balances remain completely untouched.

Feature 11: Transaction Histories & Last 4 Live Feed

Logs an unalterable receipt for all transaction attempts (detailing Type, Source, Destination, Principal Amount, Calculated Fee, Date, and status as Successful or Failed). The client homepage dashboard queries this ledger to display the user's **last 4 transactions** as an interactive activity summary.